FOOTBALL WEST

Tech, Data and AI

Presented at ECU by Aadil Vagh - Business Systems and Data Analyst

Today’s Game Plan

1. Football West & My Journey
2. Challenges in Sporting Organisations
3. Real-World Tech & AI Projects
4. Lessons Learned & What's Next

Football West



The World Game in WA

As the sole governing body for football in WA, Football West is responsible for supporting and growing the most played team sport in the state:

- from grassroots and social football to
- competitions,
- clubs and
- pathways for players, coaches and referees

Our head office is located at the Sam Kerr Football Centre with regional offices across WA.







Current Role

ME

Business Systems and Data Analyst

- Data Analytics
- Software, Systems and Web Dev
- System Architecture & AI Development
- Strategic Technology Enablement

What does that actually mean tho?

- Dashboards / Business Intelligence and Forms
- Automation tools and web pages
- Designing how systems work and talk to each other
- Helping everyone use all of the above

Pathway to Current Role



Football edits



eSports Comps



Fixtures and Table Graphics



Then automated part of my own job (the graphics) and ended up in my current role

Before we Kick Off
a FW / tech term crash course…



= Application Progamming Interface

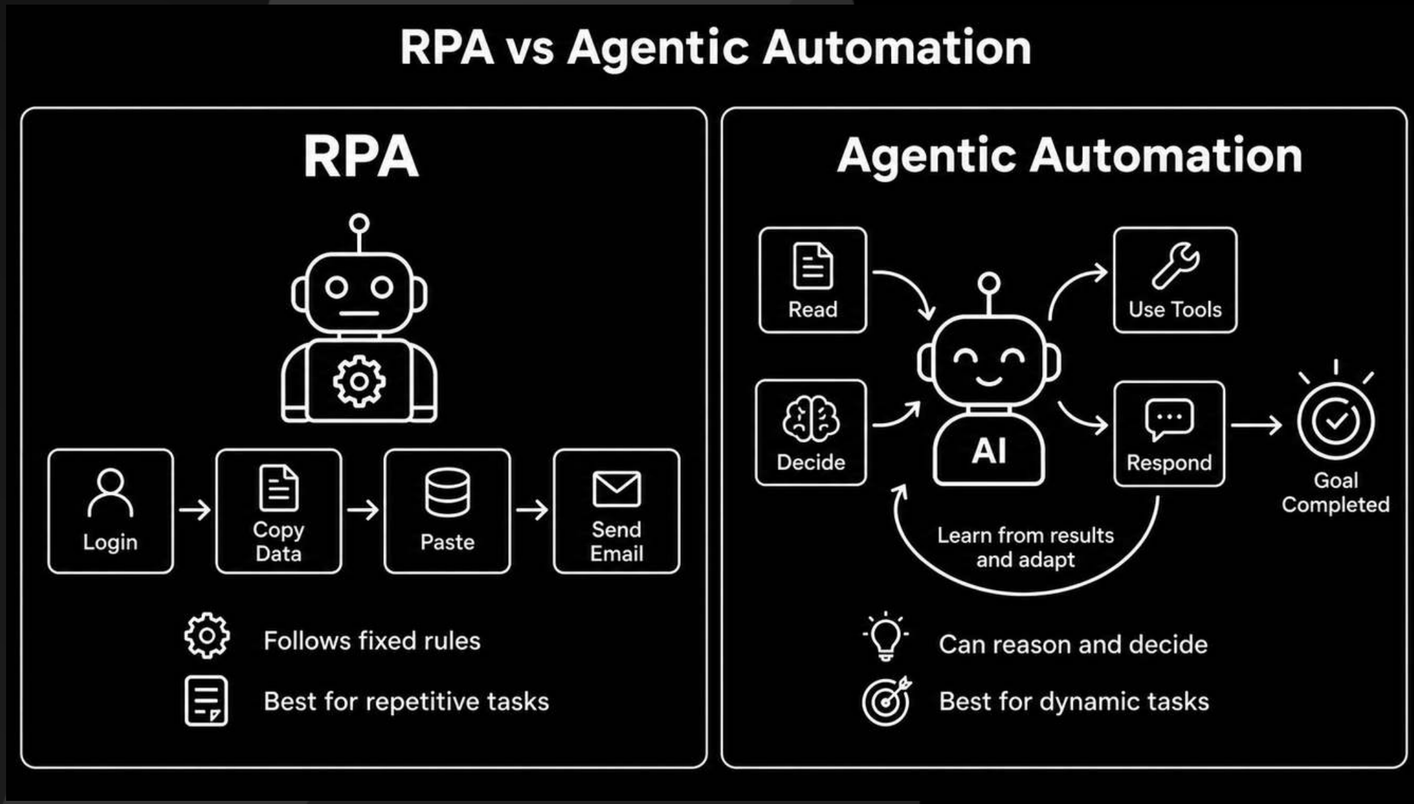
A way for different software systems to communicate, share data and automate processes.



= Python (programming language)

A widely used, versatile and beginner-friendly programming language known for its simplicity and readability.

Before we Kick Off
a FW / tech term crash course…



RPA = Robotic Process Automation - using software to automate repetitive, rule-based tasks that would otherwise be performed manually.

Again a focus of a couple projects we will discuss

Before we Kick Off
a FW / tech term crash course…



= NPL / National Premier League

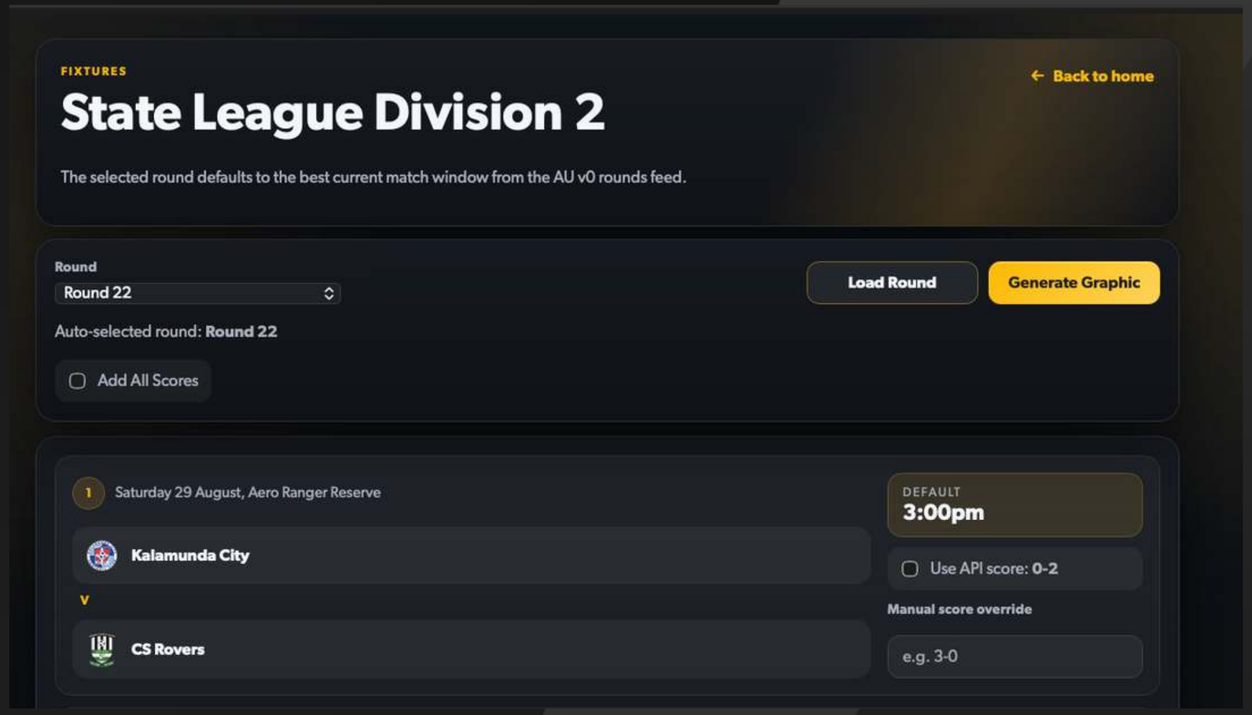
Top tier of association football played in WA
Focus of a couple of projects in this preso

CBA

= Can’t be asked

Informal slang for not having the motivation or willingness to do something.

Fixtures and Ladders Graphics Automation





The Repetitive Work

Belt Up Men's Amateur League
Premier Division

SW Phoenix	5:00pm	Jaguar	Saturday 20 May, Onside Sports Stadium
Emerald	1:00pm	Joondalup Utd	Sunday 21 May, Fraser Park
North Perth Utd	1:00pm	Kwinana Utd	Sunday 21 May, Woodville Reserve

Men's State League
Division 1

Forrestfield Utd	3:00pm	Fremantle City	Saturday 22 July, Crazy Domains Stadium
Western Knights	3:00pm	MUM FC	Saturday 22 July, Nash Field
Joondalup Utd	3:00pm	Subiaco	Saturday 22 July, Pentanet Stadium
Gwelup Croatia	3:00pm	UWA Nedlands	Saturday 22 July, Croatian Sporting Complex
Mandurah City	3:00pm	Rockingham City	Saturday 22 July, Securitas Protect Stadium

NPLWA Women's

MUM FC	7:30pm	Subiaco	Friday 12 May, Besteam Stadium
Perth SC	7:00pm	Balcatta Etna	Saturday 13 May, Dorrien Gardens
Hyundai NTC	3:00pm	Curtin University	

HYUNDAI COMMUNITY FOOTBALL RAFFLE ROUND
NPLWA Men's

Stirling Macedonia	3:00pm	Perth SC	Saturday 19 August, Macedonia Park
Floreat Athena	3:00pm	Bayswater City	Saturday 19 August, E & D Litis Stadium
Balcatta Etna	3:00pm	Armadale	Saturday 19 August, Grindleford Reserve
Inglewood Utd	3:00pm	Cockburn City	Saturday 19 August, Inglewood Stadium
Olympic Kingsway	3:00pm	Sorrento FC	

Men's State League
Division 2

		P	W	D	L	GD	PTS
1	Kingsley Westside	11	9	1	1	18	28
2	Morley Windmills	11	7	1	3	13	22
3	Swan Utd	11	6	3	2	10	21
4	MK White City	11	6	1	4	12	19
5	Wanneroo City	11	6	0	5	3	18
6	Curtin University	11	5	2	4	9	17
7	CS Rovers	11	5	1	5	1	16
8	Balga	11	4	2	5	-15	14
9	Ashfield	11	4	1	6	0	13

NPLWA Men's

		P	W	D	L	GD	PTS
1	Bayswater City	9	5	2	2	5	17
2	Stirling Macedonia	9	5	1	3	6	16
3	Armadale	9	5	0	4	7	15
4	Balcatta Etna	9	5	0	4	-2	15
5	Perth RedStar	9	4	2	3	11	14
6	Cockburn City	9	4	2	3	-4	14
7	Perth Glory	9	4	1	4	-2	13

The Problem

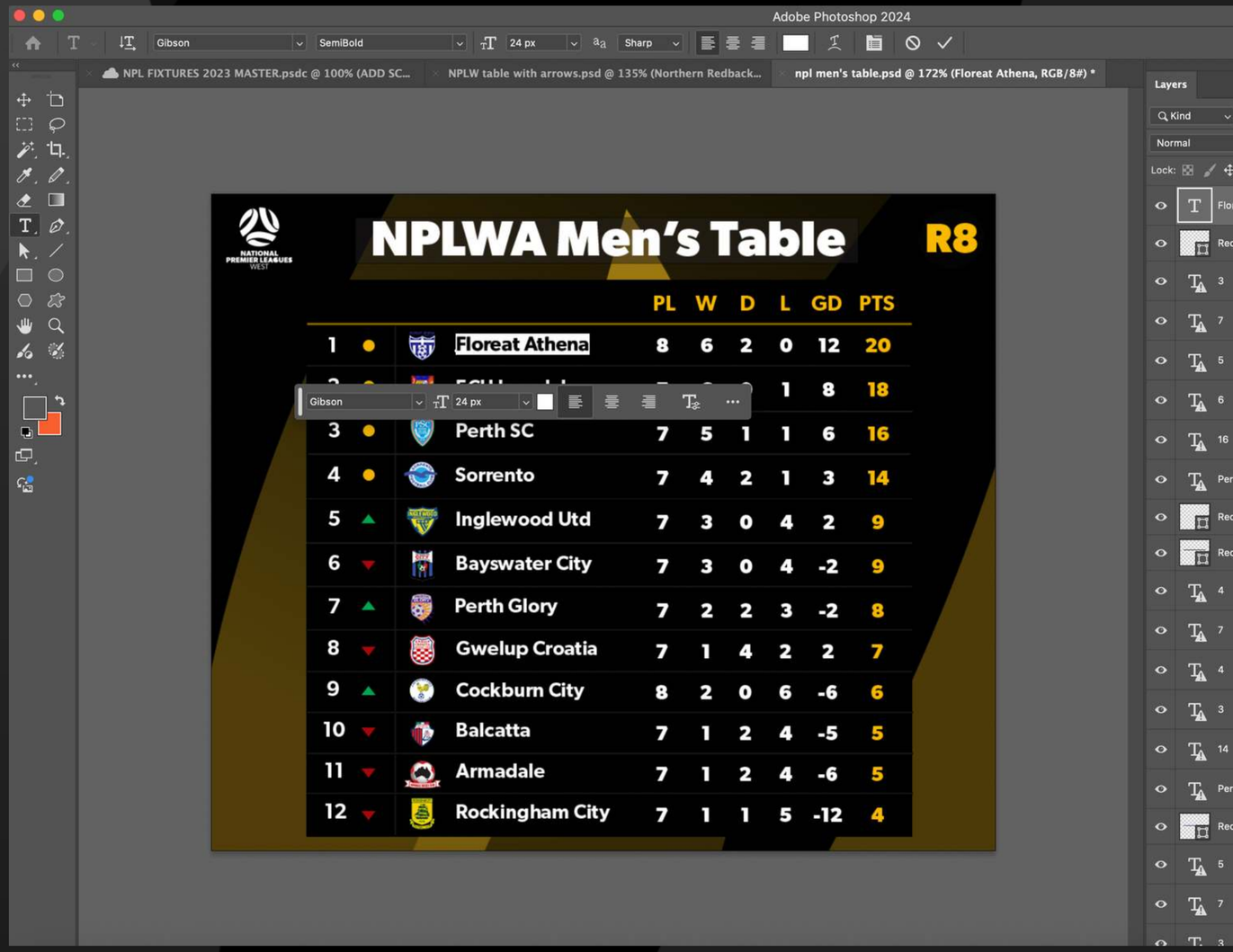
Manual fixture and ladder graphics were:

1. Time-consuming
2. Repetitive
3. Easy to get wrong

Up to about 500 manual changes per week

Even a 1% error rate = 99.6% chance of at least one error per week

P(error) = 1 - (1 - p)^ n

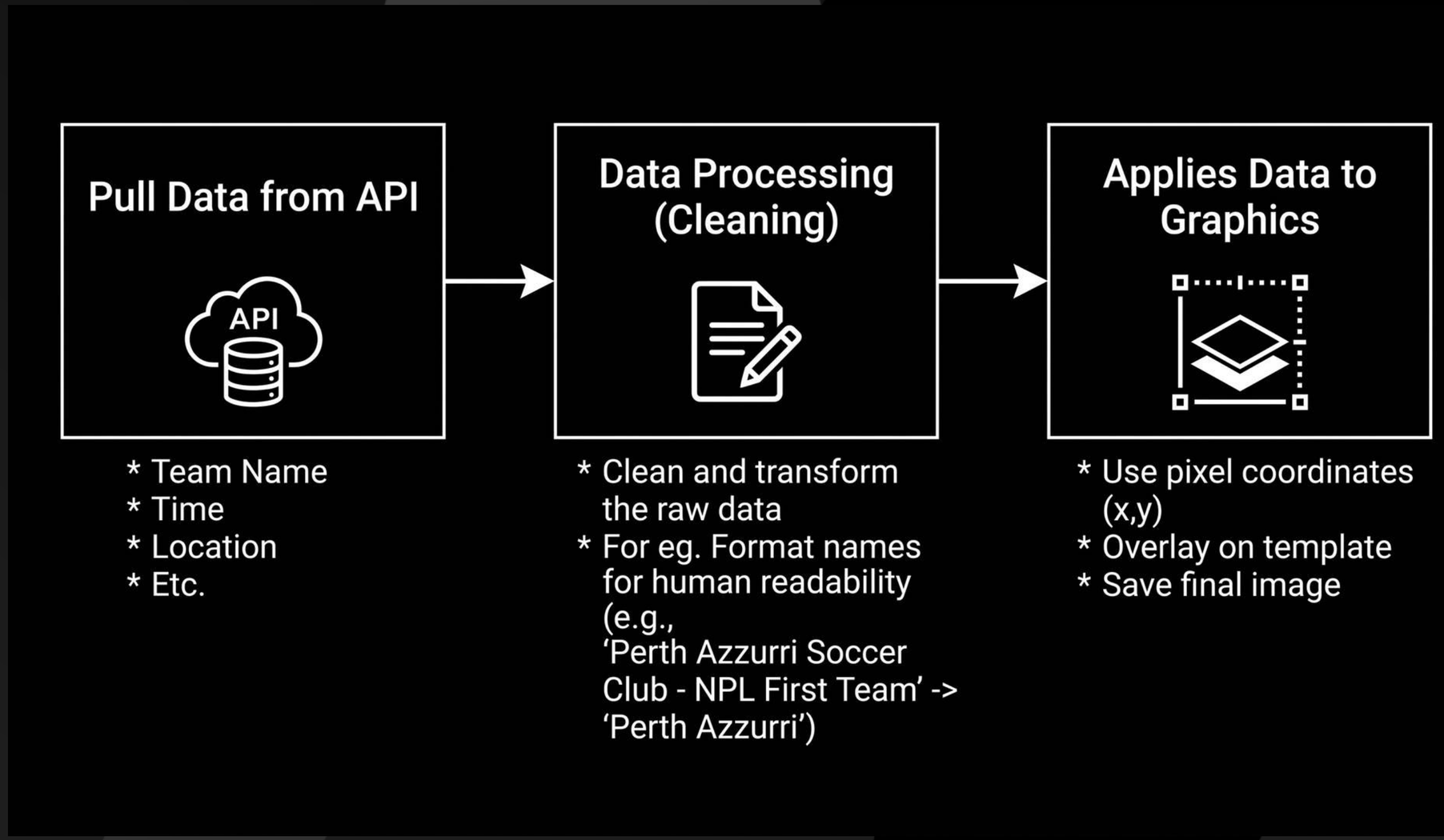


The Solution

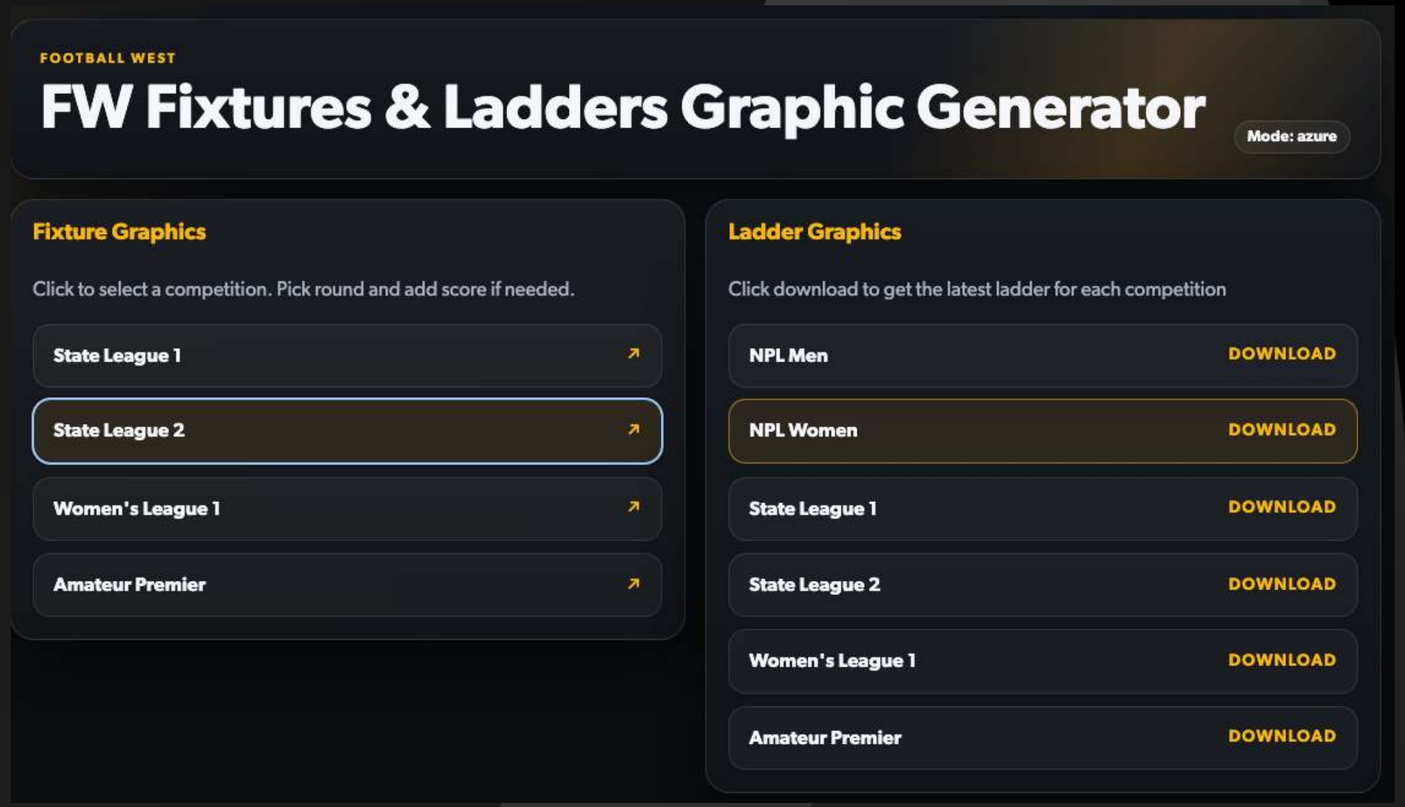
get someone/thing else to do the work for me

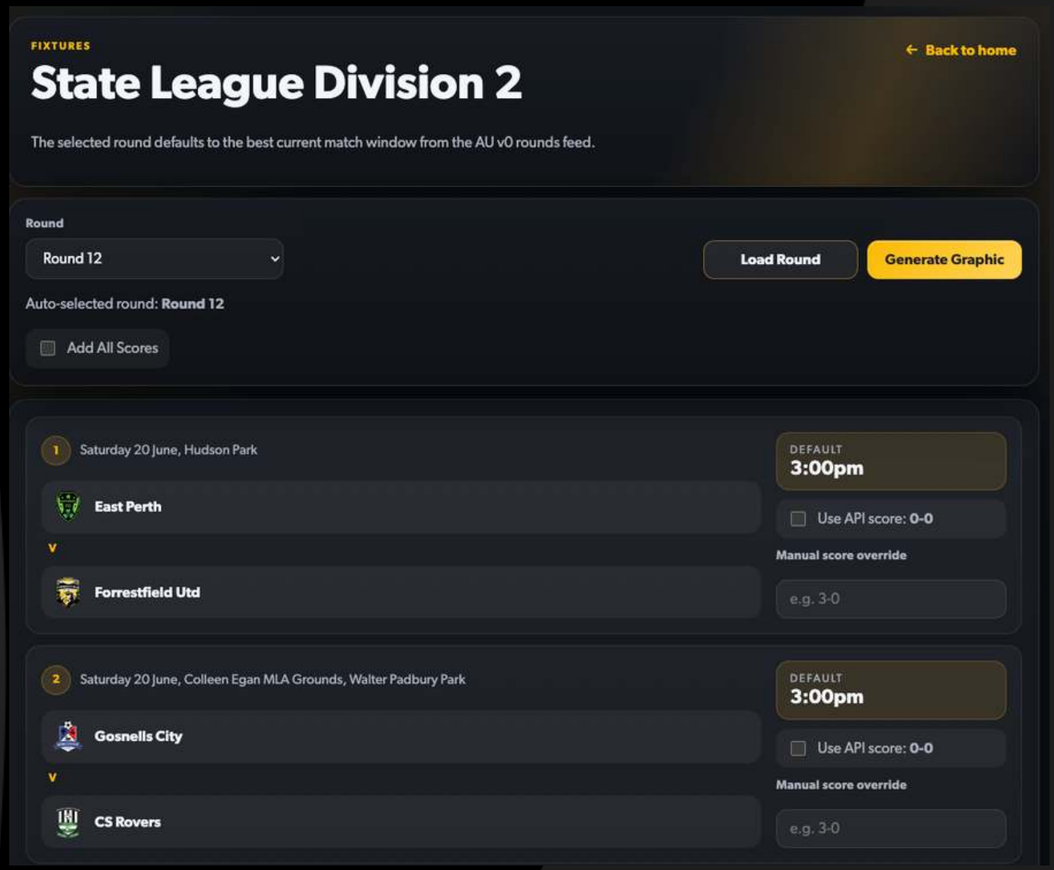


The Solution



Cloud based portal





Benefits

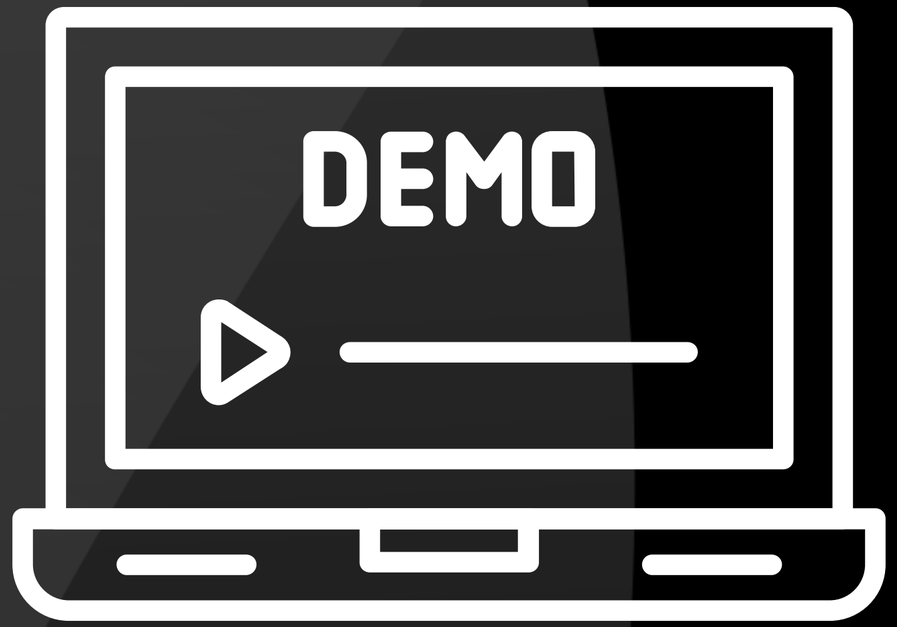
What used to take 4-5 hours / week now takes less than a minute

Frees up time for other tasks such as those that need more creativity etc

Error free*

Get invited to talk about it at guest lectures

Simpler ways to achieve this



Simpler ways to achieve this



You've just been hired in the marketing team for the brand new State Sporting Org - BUBBLE SOCCER WA

The CEO wants you to post fixtures and ladders for about 10+ divisions every week of the season

How would you try and automate this?

LLMs are inconsistent



Bubble Soccer WA - DIV 1

#	TEAM	P	W	D	L	GD	PTS
1	Bubble Bros	12	10	1	1	24	31
2	Ball Bumpers	12	9	2	1	18	29
3	Air Force	12	8	2	2	15	26
4	Bump & Run	12	7	3	2	12	24
5	Bubble FC	12	6	2	4	5	20
6	Inflatable United	12	5	2	5	-2	17
7	The Bubbliers	12	4	3	5	-6	15
8	Bounce Patrol	12	3	2	7	-11	11
9	Bubble Trouble	12	2	1	9	-18	7
10	Last Man Bouncing	12	0	2	10	-37	2

ChatGPT medium thinking model

Completely made up team names



Bubble Soccer WA - DIV 1

#	Team Name	P	W	D	L	GD	PTS
1	Perth Pythons	10	8	1	1	22	25
2	Fremantle Falcons	10	7	2	1	18	23
3	Joondalup Jets	10	6	2	2	11	20
4	Rockingham Rhinos	10	5	2	3	6	17
5	Mandurah Mariners	10	4	3	3	2	15
6	Subiaco Stingrays	10	4	1	5	-3	13
7	Midland Magpies	10	3	2	5	-7	11
8	Bunbury Bulls	10	2	3	5	-9	9
9	Armadale Avalanche	10	2	1	7	-16	7
10	Wanneroo Wolves	10	1	1	8	-24	4

Gemini Pro

Not terrible but doesnt look amazing

Scripted vs AI-Generated Image Outputs

A script is repeatble and modifiable
whereas LLMs are a bit black of a black box, and as seen on the last slide produce inconsistent and unpredictable results

```
#PASTE LADDER DATA
t1 = ImageDraw.Draw(img)
t1.text((276.5, 250), joined1 , align='left', font = font_semibold, fill =('#ffffff'))
t1.text((612, 277), joined2 , align='center', font = font_semibold, fill =('#ffffff'), anchor = 'ms')
t1.text((677, 277), joined3 , align='center', font = font_semibold, fill =('#ffffff'), anchor = 'ms')
t1.text((740, 277), joined4 , align='center', font = font_semibold, fill =('#ffffff'), anchor = 'ms')
t1.text((800, 277), joined5 , align='center', font = font_semibold, fill =('#ffffff'), anchor = 'ms')
t1.text((860, 277), joined6 , align='center', font = font_semibold, fill =('#ffffff'), anchor = 'ms')
t1.text((940, 277), joined7 , align='center', font = font_bold, fill =('#fbba00'), anchor = 'ms')
```

In the Python script the paste locations, fonts and colours can be changed relatively easily

Cybersecurity



Never paste API keys or passwords directly in your code.



Use a sandbox environment for testing, with dummy data.



Don’t blindly trust AI-generated code. Review, understand and test it first.



If you’re going to deploy code publicly or make it accessible to others, and you’re unsure, always have it reviewed and tested by a cybersecurity expert first.

Full-Time

Start with the Problem

Not every problem needs AI or an expensive solution.

Tech is just Half the Battle

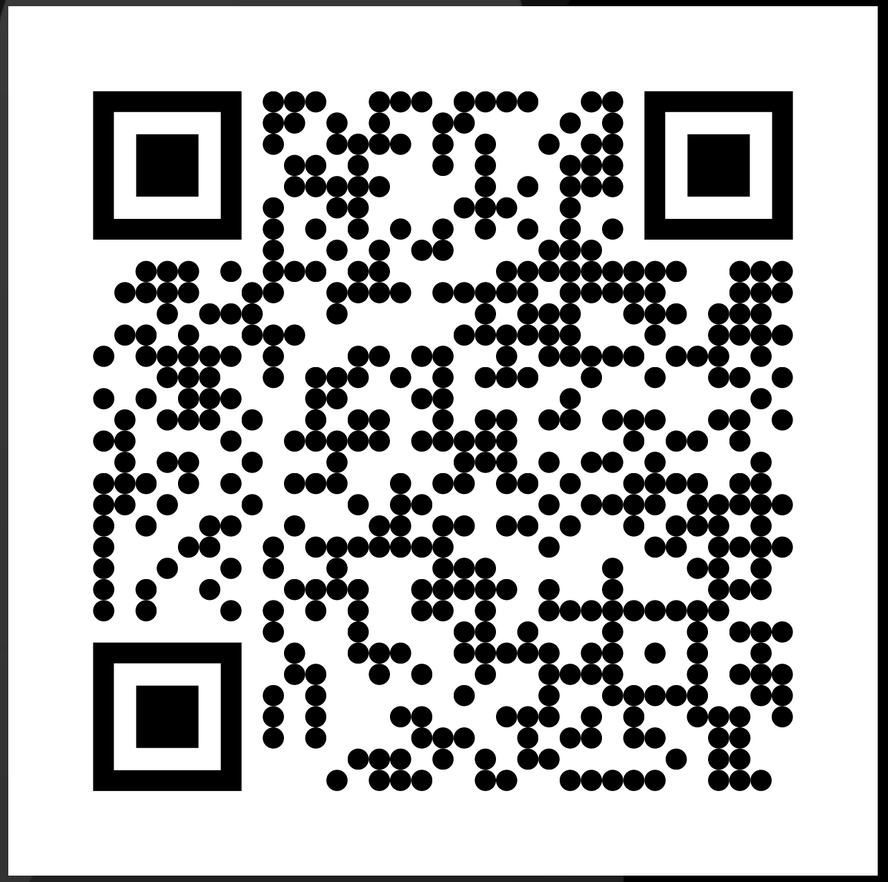
Getting people to trust and actually use it is just as important.

Start Small & Experiment

Basically if you have an idea, just give it a crack. You don't need a huge team or budget to make a meaningful impact.

Automate the boring stuff. Focus on what matters.

That’s it
Questions?



QR to my Linkedin if youre keen to connect